ADSA Syllabus

Theory: 45 Lecture Plan

Pedagogical Tools (all lectures): Activity, Case Study, Code, Flipped Classes, Infographics, Instructor Lead Workshop, PPT, Reports, Simulation, Video Lecture

Text/Reference Books (all lectures):

- T-Introduction to Algorithms — Cormen, Leiserson, Rivest, Stein (4th Ed, MIT Press, 2022)
- T-Guide to Competitive Programming — Antti Laaksonen (3rd Ed, Springer, 2024)
- T-The Algorithm Design Manual — Steven S. Skiena (3rd Ed, Springer, 2020)
- R-Competitive Programming — Halim, Halim, Effendy (4th Ed, Lulu, 2020)

Unit 1 — Fundamentals of Data Structures & Sorting (15 Lectures)

Chapter 1.1: Fundamentals of Data Structures & Arrays

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
1	L1	Fundamentals of Data Structures & Array	Concept of data, information, and introduction to data structures	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Video Lecture	CO1	BT4
1	L2	Fundamentals of Data Structures & Array	Algorithm complexity — time and space; time-space tradeoff	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Case Study, PPT, Code	CO1	BT4
1	L3	Fundamentals of Data Structures & Array	Asymptotic notations — Big-O, Big-Ω, Big-Θ with examples	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO1	BT4

1	L4	Fundamentals of Data Structures & Array	Linear arrays — representation and traversal	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO1, CO2	BT4
1	L5	Fundamentals of Data Structures & Array	Insertion and deletion in arrays	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2	BT3
1	L6	Fundamentals of Data Structures & Array	Searching — linear search and binary search with complexity analysis	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Case Study	CO1, CO5	BT4

Chapter 1.2: Sorting Techniques

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
1	L7	Sorting Techniques	Introduction to sorting; analysis of basic sorting techniques	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Video Lecture	CO1, CO5	BT4
1	L8	Sorting Techniques	Bubble sort — algorithm and complexity analysis	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO1, CO5	BT4

1	L9	Sorting Techniques	Quick sort — algorithm and complexity analysis	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO1, CO5	BT4
1	L10	Sorting Techniques	Merge sort — merging arrays and complexity analysis	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO1, CO5	BT4
1	L11	Sorting Techniques	Multi-dimensional arrays — representation and applications	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2	BT3
1	L12	Sorting Techniques	Recap and comparative analysis of sorting techniques	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Case Study, Code, Reports	CO1, CO5	BT4

Chapter 1.3: Pointers, Records & Advanced Arrays

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
1	L13	Pointers, Records & Advanced Arrays	Pointers and pointer arrays	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Video Lecture	CO2	BT3

1	L14	Pointers, Records & Advanced Arrays	Records — structure and memory representation; parallel arrays	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2	BT3
1	L15	Pointers, Records & Advanced Arrays	Sparse matrices — representation and storage techniques	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO4	BT3

Self Study (Unit 1): Amortized Analysis, Advanced Recurrence Relations, Threaded Binary Trees, Red-Black Trees, B-Trees and B+ Trees, Splay Trees, Huffman Trees, Decision Trees, Space-Time Trade-offs

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Unit 2 — Linked Lists, Stacks & Queues (14 Lectures)

Chapter 2.1: Linked Lists

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
2	L16	Linked Lists	Linear linked list — representation in memory	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2	BT3
2	L17	Linked Lists	Traversal and searching in a linked list	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO5	BT3

2	L18	Linked Lists	Insertion and deletion in a linked list	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2	BT3
2	L19	Linked Lists	Header linked list and doubly linked list	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2	BT3
2	L20	Linked Lists	Circular linked list and circular doubly linked list	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2	BT3
2	L21	Linked Lists	Operations on doubly linked list; complexity analysis and applications	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Case Study, Code, Reports	CO1, CO2	BT4

Chapter 2.2: Stacks & Recursion

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
2	L22	Stacks & Recursion	Stack — basic terminology, sequential and linked representations	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2	BT3

2	L23	Stacks & Recursion	Stack operations — PUSH & POP; parenthesis matching	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO3	BT5
2	L24	Stacks & Recursion	Evaluation of postfix expressions	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Case Study	CO2, CO3	BT5
2	L25	Stacks & Recursion	Infix to postfix conversion	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO3	BT5
2	L26	Stacks & Recursion	Recursion — principles, importance, and implementation	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Video Lecture	CO2, CO3	BT5

Chapter 2.3: Queues

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
2	L27	Queues	Queues — introduction and operations	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2	BT3

2	L28	Queues	Double ended queue (Deque)	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2	BT3
2	L29	Queues	Priority queue — concept and applications	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Case Study, Code, Reports	CO2, CO4	BT3

Self Study (Unit 2): Topological Sorting (in-depth), Strongly Connected Components, Floyd-Warshall Algorithm, Ford-Fulkerson Network Flow, Advanced Graph Coloring, Perfect Hashing, Double Hashing, Extendible Hashing, File Indexing, B+ Tree Indexing in Databases

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Unit 3 — Graphs, Trees, Hashing & File Organization (16 Lectures)

Chapter 3.1: Graphs & Binary Trees

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
3	L30	Graphs & Binary Trees	Graph theory — terminology, adjacency matrix, path matrix	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Infographics, Video Lecture	CO2	BT3
3	L31	Graphs & Binary Trees	Graph traversal — BFS	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO3	BT5

3	L32	Graphs & Binary Trees	Graph traversal — DFS and comparison with BFS	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Case Study	CO2, CO3, CO5	BT5
3	L33	Graphs & Binary Trees	Binary trees — terminology, representation in memory	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Infographics, Code	CO2	BT3
3	L34	Graphs & Binary Trees	Binary tree traversal algorithms (using stacks)	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO3	BT5
3	L35	Graphs & Binary Trees	Shortest path — Dijkstra's Algorithm	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Case Study	CO3, CO4	BT5

Chapter 3.2: Binary Search Trees & Advanced Trees

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
3	L36	Binary Search Trees & Advanced Trees	Shortest path — Bellman-Ford Algorithm	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Case Study	CO3, CO4	BT5

3	L37	Binary Search Trees & Advanced Trees	Binary search trees — search, insert, and delete	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO4	BT6
3	L38	Binary Search Trees & Advanced Trees	AVL search trees — rotations and balancing	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO4	BT6
3	L39	Binary Search Trees & Advanced Trees	B-Trees — structure and operations	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2, CO4	BT6
3	L40	Binary Search Trees & Advanced Trees	Heaps — min-heap, max-heap	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO3	BT5
3	L41	Binary Search Trees & Advanced Trees	Heap sort and comparative analysis of tree structures	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Case Study, Code, Reports	CO3, CO5	BT5

Chapter 3.3: Hashing & File Organization

Unit No	Lecture No	Chapter Name	Topic	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level

3	L42	Hashing & File Organization	Hash tables and hash functions	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Infographics	CO2	BT3
3	L43	Hashing & File Organization	Collision resolution strategies (chaining, linear probing)	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Code, Simulation	CO2, CO3	BT5
3	L44	Hashing & File Organization	File organization — sequential, relative, index sequential, inverted file	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, PPT, Infographics, Video Lecture	CO2, CO4	BT6
3	L45	Hashing & File Organization	Recap — designing efficient data systems (real-world applications)	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Case Study, Code, Reports	CO4, CO5	BT6

Self Study (Unit 3): Randomized Algorithms, Karatsuba Algorithm, Closest Pair Problem, Backtracking vs Divide & Conquer, Greedy vs DP comparison, 0/1 Knapsack (DP), Disjoint Set (Union-Find), Approximation Algorithms, Branch and Bound

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Practical: 15 Experiment Plan

Pedagogical Tools (all experiments): Activity, Case Study, Code, Flipped Classes, Infographics, Instructor Lead Workshop, PPT, Reports, Simulation, Video Lecture

Text/Reference Books (all experiments):

- T-Introduction to Algorithms — Cormen, Leiserson, Rivest, Stein (4th Ed, MIT Press, 2022)

- T-Guide to Competitive Programming — Antti Laaksonen (3rd Ed, Springer, 2024)
- T-The Algorithm Design Manual — Steven S. Skiena (3rd Ed, Springer, 2020)
- R-Competitive Programming — Halim, Halim, Effendy (4th Ed, Lulu, 2020)

Unit No	Exp No	Experiment Name	LeetCode Link	Text/Reference Books	Pedagogical Tools	Mapped CO(s)	BT Level
1	E1	Search in Rotated Sorted Array (#33) — Binary Search	LC #33	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO1, CO5	BT4
1	E2	Sort an Array (#912) — Merge Sort / Quick Sort	LC #912	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO1, CO5	BT4
1	E3	Group Anagrams (#49) — Hash Table & Collision Resolution	LC #49	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Simulation, Reports	CO2, CO3	BT5
1	E4	Kth Largest Element in an Array (#215) — Quick Sort / Divide & Conquer	LC #215	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO3, CO5	BT5
2	E5	Swap Nodes in Pairs (#24) — Linked List Operations	LC #24	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Simulation, Reports	CO2	BT3

2	E6	Evaluate Reverse Polish Notation (#150) — Stack & Postfix	LC #150	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO2, CO3	BT5
2	E7	Number of Islands (#200) — BFS & DFS Graph Traversal	LC #200	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO2, CO3	BT5
3	E8	Network Delay Time (#743) — Dijkstra's Shortest Path	LC #743	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO3, CO4	BT5
3	E9	Non-overlapping Intervals (#435) — Greedy / Knapsack	LC #435	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Reports	CO3, CO4	BT5
3	E10	Course Schedule (#207) — Topological Sorting	LC #207	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO3, CO4	BT5
3	E11	Maximum Depth of Binary Tree (#104) — DFS / BFS on Trees	LC #104	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO2, CO3	BT5

3	E12	Number of Islands (#200) — BFS & DFS Graph Traversal	LC #200	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO2, CO3	BT5
3	E13	Binary Tree Level Order Traversal (#102) — BFS / Queue	LC #102	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Simulation	CO2, CO3	BT5
3	E14	Valid Anagram (#242) — Hashing & Frequency Count	LC #242	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Simulation, Reports	CO2, CO3	BT5
3	E15	Combination Sum (#39) — Backtracking / Recursion	LC #39	T-Introduction to Algorithms, T-Guide to Competitive Programming, T-The Algorithm Design Manual, R-Competitive Programming	Activity, Code, Case Study, Reports	CO3, CO4	BT5